Hourly price fluctuation analysis

A detailed mathematical and empirical report on CVX, NVDA and PG under Iran-US war exposure

MAC-F244 - Stochastic Calculus and Application to Finance

Research question. How do three structurally different stocks absorb geopolitical risk, and which forecasting models best predict their out-of-sample hourly price paths? The report studies Chevron (CVX), NVIDIA (NVDA) and Procter & Gamble (PG) using hourly market data, oil benchmarks and a defence proxy.

Data window. In-sample: 16 March to 15 April 2026, with 153 hourly returns across 22 trading sessions. Out-of-sample: 16 April to 24 April 2026, with 49 hourly bars across 7 trading sessions. Auxiliary proxies are Brent crude, WTI crude and Lockheed Martin. The files used are the cached CSVs and figures supplied in the submission folder.

Ticker	Economic role in the design
CVX	Oil-direct name: should absorb oil and Hormuz news, especially through overnight open gaps.
NVDA	High-beta risk asset: should absorb war news through risk sentiment rather than direct oil cash flows.
PG	Defensive consumer staple: should be relatively insulated from oil-war shocks.

Generated from the supplied analysis folder. All numerical tables and plots come from `task2_pipeline.py` outputs.

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How to read this report

This rewritten version gives the mathematical reasoning behind every major empirical step. Instead of only reporting which model won or which stock looked more exposed, it explains how each statistic is constructed, what assumption sits underneath it, and what inference can safely be drawn from it. The aim is to make the report viva-ready: every table, equation and figure is tied to an interpretation.

The report follows a simple logic. First, prices are converted into returns because returns are closer to stationary than price levels. Next, the return distributions are tested for volatility, skewness, tail risk and normality. Then several forecasting models are compared on a held-out test window. Finally, geopolitical exposure is decomposed into three channels: intraday co-movement, overnight oil transmission and jump/event response.

1 Executive summary

The main empirical result is not that one stock is simply "more affected" than another. The sharper result is that each stock absorbs war risk through a different mathematical channel. CVX is the cleanest oil-direct name through overnight gaps. NVDA is the strongest continuous risk-sentiment mover. PG behaves like the defensive control group.

CVX has the most violent distributional tail: hourly skewness is -2.54 and excess kurtosis is 14.31. This is the signature of a jump-sensitive oil major rather than a calm Gaussian diffusion. NVDA has a positive drift over the training window and responds strongly to de-escalation events. PG has the lowest volatility and the weakest relationship with oil proxies.

For forecasting, no single model dominates across all names. Naive/ARIMA wins for CVX because the test window behaves like a random walk after a training-window downtrend. GBM wins for NVDA because its positive in-sample drift persists into the test window. ARIMA$(2, 1, 2)$ wins for PG, showing that the low-volatility staple has more exploitable serial structure than the two noisier names.

For war exposure, the composite score ranks NVDA first and CVX second, but this needs a careful reading. CVX wins the cleanest oil-direct channel: overnight Brent return explains about 51% of the variance in CVX's open-gap return. NVDA wins the broader composite because its risk-off/risk-on sensitivity shows up continuously during market hours and in event-day reactions.

2 Data design and mathematical setup

2.1 Choice of assets

The three-stock design is deliberately cross-sectional. It does not merely ask whether one asset moved. It asks whether the same geopolitical shock leaves different mathematical fingerprints on different balance sheets and investor narratives.

- CVX is exposed through commodity cash-flow expectations. If oil prices jump because of supply disruption, CVX can benefit.
- NVDA is exposed through risk appetite. Oil-war escalation can raise inflation and uncertainty, which can pressure high-duration growth stocks.
- PG is a defensive control. If the method is sensible, PG should show the weakest geopolitical linkage.

Let S_t denote the hourly close price. The simple return and log return are

$$r_t = \frac{S_t - S_{t-1}}{S_{t-1}}, \qquad \ell_t = \log S_t - \log S_{t-1}.$$

Simple returns are used for most empirical correlations because they are directly interpretable in percentage terms. Log returns are used in the GBM model because the stochastic differential equation is naturally written in log-price form.

2.2 Why hourly data matters

Daily data can hide the mechanism. Oil trades almost continuously, while US equities have a regular trading session. A Brent shock can happen when NYSE is closed. CVX may then absorb the shock in the first bar after the open, not continuously across all intraday bars. This is why the report separates intraday, overnight and event channels.

2.3 The probability model behind the notation

At each hourly timestamp, the observed price is treated as one realisation of a random process. Formally, the price sequence is written as

$$\{S_t : t = 0, 1, 2, \ldots, T\},$$

where S_t is the traded stock price at hour t. The return sequence is then a transformed process,

$$\{r_t : t = 1, 2, \ldots, T\}, \qquad r_t = \frac{S_t - S_{t-1}}{S_{t-1}}.$$

This transformation is central. Prices of equities often have trends and unit-root-like behaviour, while returns are usually closer to stable fluctuations around a mean. That is why most tests in the report are applied to returns rather than raw prices.

For small hourly moves, simple and log returns are almost identical because

$$\log(1 + r_t) = r_t - \frac{r_t^2}{2} + \frac{r_t^3}{3} - \cdots \approx r_t.$$

The approximation is good when $|r_t|$ is small. However, log returns are more convenient for continuous-time models because multi-period log returns add:

$$\log \frac{S_T}{S_0} = \sum_{t=1}^{T} \log \frac{S_t}{S_{t-1}}.$$

Simple returns do not add in this way; they compound multiplicatively. This is why the empirical tables use simple returns for direct percentage interpretation, while GBM uses log returns for mathematical consistency.

2.4 Train-test split and why it matters

The analysis uses an in-sample window for estimating model parameters and a later out-of-sample window for testing forecasts. This avoids a common error: fitting a model to all available data and then claiming it predicts the same data. If $\mathcal{F}_T$ represents the information known at the end of the training window, a valid forecast must satisfy

$$\hat{S}_{T+h} = E[S_{T+h} \mid \mathcal{F}_T]$$

or at least be computable using only information contained in $\mathcal{F}_T$. The recursive machine-learning forecasts follow this rule: once the test window begins, the model uses its own past predictions to form future lagged features instead of secretly using actual future prices.

2.5 What counts as an inference

A numerical result becomes an inference only after it is connected to the economic mechanism. For example, a positive Brent-CVX overnight correlation means more than co-movement. It suggests that oil information arriving when the US equity market is closed is incorporated into CVX at the next open. Similarly, a negative Brent-NVDA correlation does not mean NVDA sells oil. It indicates that oil shocks may operate through inflation pressure, uncertainty and risk appetite.

3 Distributional analysis

3.1 Sample moments and trend statistics

The first mathematical object is the empirical distribution of hourly returns. A distribution is not described only by its average. For financial returns, the average is often small, while volatility, skewness and kurtosis carry most of the information about risk. The sample mean, standard deviation, skewness and excess kurtosis are estimated as

$$\hat{\mu} = \frac{1}{n}\sum_{t=1}^{n} r_t, \qquad \hat{\sigma}^2 = \frac{1}{n-1}\sum_{t=1}^{n}(r_t - \hat{\mu})^2,$$

$$\hat{\gamma}_1 = \frac{\frac{1}{n}\sum_t (r_t - \hat{\mu})^3}{\hat{\sigma}^3}, \qquad \hat{\gamma}_2 = \frac{\frac{1}{n}\sum_t (r_t - \hat{\mu})^4}{\hat{\sigma}^4} - 3.$$

Annualised volatility is computed as $\hat{\sigma}\sqrt{6.5 \times 252}$ because there are approximately 6.5 trading hours per US equity trading day and 252 trading days per year.

Table 1: Core distributional and trend statistics

Ticker	n	Mean (%/hr)	Std (%/hr)	Ann. vol (%)	Total return (%)	Skew	Ex. kurt.	Trend β	R^2
CVX	153	-0.04	0.89	35.93	-6.35	-2.54	14.31	-0.107	0.369
NVDA	153	0.05	0.77	31.07	8.04	1.06	5.03	0.081	0.216
PG	153	-0.03	0.40	16.15	-5.27	-0.15	5.42	-0.032	0.294

The table gives the first major inference. CVX is the most tail-risk-heavy name. Its standard deviation is high, its skew is sharply negative, and its excess kurtosis is almost three times NVDA's. In plain language, CVX did not merely fluctuate. It carried a crash-like left tail during the sample.

NVDA has a different signature. Its mean hourly return is positive and the total in-sample return is about 8%. This matters for forecasting because models that preserve a positive drift, especially GBM and drift extrapolation, have an advantage if that drift continues.

PG is the defensive benchmark. Its annualised volatility is almost half of CVX's. The negative total return shows that it was not immune to market pressure, but its lower variance and weaker oil linkage support the view that it is comparatively insulated.

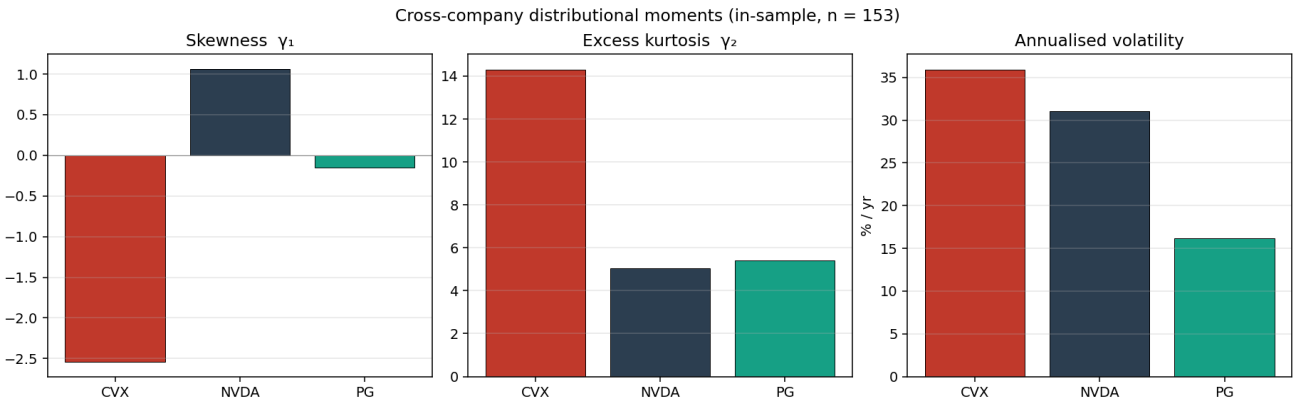


Figure 1: Cross-company comparison of skewness, excess kurtosis and annualised volatility.

3.2 Normality and stationarity

The Jarque-Bera test combines skewness and kurtosis:

$$JB = \frac{n}{6}\left(\hat{\gamma}_1^2 + \frac{\hat{\gamma}_2^2}{4}\right).$$

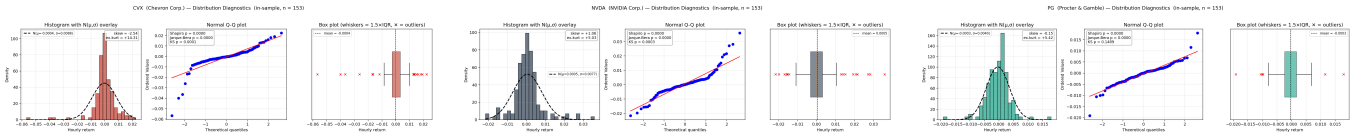
A small p-value rejects the Gaussian assumption. The Augmented Dickey-Fuller test is used to separate non-stationary price levels from more stable return series.

Table 2: Normality and stationarity diagnostics

Ticker	Shapiro p	JB p	KS p	ADF price p	ADF return p
CVX	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$	0.8746	0.1199
NVDA	$< 10^{-4}$	$< 10^{-4}$	0.0003	0.9763	$< 10^{-4}$
PG	$< 10^{-4}$	$< 10^{-4}$	0.1489	0.1242	$< 10^{-4}$

The normality tests reject Gaussian returns for all three names under Shapiro-Wilk and Jarque-Bera. This is not a technical footnote. It directly affects model choice. A pure Gaussian model underprices jump risk in CVX because its observed left tail is far too large for a normal distribution with the same variance.

The price series behave like non-stationary levels, which is normal for equity prices over short windows. The return series are more stable for NVDA and PG. CVX returns are borderline under the ADF test, which is consistent with the heavy-tail shock contaminating a short sample.



(a) CVX **(b)** NVDA **(c)** PG

Figure 2: Distribution panels: histogram with normal overlay, Q-Q plot and box plot.

4 Forecasting methodology

4.1 Forecast target

The forecast target is the hourly price trajectory over the test window:

$$\hat{S}_{T+1}, \hat{S}_{T+2}, \ldots, \hat{S}_{T+h}, \qquad h = 49.$$

This is more informative than trying to predict raw hourly returns directly. Hourly returns are noisy and close to a martingale-difference sequence. Price trajectories allow drift, mean reversion and recursive model errors to be compared clearly.

4.2 Models used

Eight forecasting models are compared:

1. Naive random walk: $\hat{S}_{T+k} = S_T$.
2. Constant drift: $\hat{S}_{T+k} = S_T + \delta k$, where $\delta = (S_T - S_0)/(N-1)$.
3. ARIMA(p, d, q) selected by AIC over a small grid.
4. Geometric Brownian Motion with Monte Carlo path averaging.
5. Ridge regression on lagged returns, rolling moments, price lags and time features.
6. Random Forest on the same feature set.
7. Gradient Boosting on the same feature set.
8. Validation-weighted ensemble of the best three validation models.

The ARIMA model is

$$\Phi(L)(1-L)^d S_t = c + \Theta(L)\varepsilon_t,$$

where L is the lag operator. The GBM model is

$$\frac{dS_t}{S_t} = \mu\, dt + \sigma\, dW_t,$$

so Ito's lemma gives

$$d \log S_t = \left(\mu - \frac{1}{2}\sigma^2\right) dt + \sigma\, dW_t.$$

The simulation therefore uses

$$S_{T+k} = S_T \exp\left(\sum_{j=1}^{k}\left(\hat{\mu} - \frac{1}{2}\hat{\sigma}^2\right) + \hat{\sigma} Z_j\right), \qquad Z_j \sim \mathcal{N}(0,1).$$

4.3 Why recursive forecasting is important

The machine-learning models are not allowed to peek at future prices. Once the first test return is predicted, it is converted into a predicted price. That predicted price becomes part of the feature history for the next step. This recursive design is stricter and more realistic than one-step testing with actual future lags.

4.4 Why the forecast is made on prices, not only returns

A one-hour return forecast answers a narrow question: what is the next small percentage move? A price-path forecast answers a broader question: where will the asset be over the full test window? This matters because small directional errors can compound into large level errors.

If a model predicts returns $\hat{r}_{T+1}, \ldots, \hat{r}_{T+h}$, the implied price path is

$$\hat{S}_{T+k} = S_T \prod_{j=1}^{k}(1 + \hat{r}_{T+j}).$$

For log-return forecasts, it becomes

$$\hat{S}_{T+k} = S_T \exp\left(\sum_{j=1}^{k} \hat{\ell}_{T+j}\right).$$

Thus, forecasting is a compounding problem. A model can have small one-step return errors but still drift badly if its average bias has the wrong sign.

4.5 Naive random walk as a benchmark

The naive model is

$$\hat{S}_{T+k} = S_T.$$

This looks too simple, but it is a powerful benchmark because liquid stock prices are difficult to predict. If a complex model cannot beat the last observed price, the added complexity is not useful. Mathematically, the naive model corresponds to the conditional expectation of a martingale:

$$E[S_{T+k} \mid \mathcal{F}_T] = S_T.$$

This is why the naive model should always be included in a short-horizon finance study.

4.6 Drift model and its weakness

The drift model estimates a constant linear movement from the training window:

$$\hat{\delta} = \frac{S_T - S_0}{T}, \qquad \hat{S}_{T+k} = S_T + k\hat{\delta}.$$

Its weakness is regime dependence. If the training window contains a downtrend but the test window flattens, the model keeps pushing the forecast down. This explains why drift performs poorly for CVX and PG in RMSE despite being easy to interpret.

4.7 ARIMA in detail

An ARIMA(p, d, q) model has three pieces. The d term differences the series to remove non-stationarity. The AR part uses past values. The MA part uses past forecast errors. In lag notation,

$$\Phi(L)(1 - L)^d S_t = c + \Theta(L)\varepsilon_t,$$

where

$$\Phi(L) = 1 - \phi_1 L - \cdots - \phi_p L^p, \qquad \Theta(L) = 1 + \theta_1 L + \cdots + \theta_q L^q.$$

For CVX and NVDA, the selected order is ARIMA$(0, 1, 0)$, so the model reduces to

$$\Delta S_t = \varepsilon_t.$$

That is exactly a random walk. For PG, ARIMA$(2, 1, 2)$ means the differenced price has two autoregressive lags and two moving-average error lags. This is why PG is the only stock where ARIMA adds structure beyond the naive forecast.

The order is selected using AIC:

$$AIC = 2k - 2\log(\hat{L}),$$

where k is the number of estimated parameters and $\hat{L}$ is the maximised likelihood. AIC rewards fit through the likelihood term but penalises unnecessary parameters through $2k$.

4.8 GBM and Ito's lemma

GBM assumes the continuous-time stochastic differential equation

$$dS_t = \mu S_t dt + \sigma S_t dW_t.$$

Here μ is the drift, σ is volatility and W_t is Brownian motion. To derive the log-price equation, apply Ito's lemma to $f(S_t) = \log S_t$. Since

$$f'(S) = \frac{1}{S}, \qquad f''(S) = -\frac{1}{S^2},$$

Ito's lemma gives

$$d\log S_t = f'(S_t)dS_t + \frac{1}{2}f''(S_t)(dS_t)^2.$$

Because $(dW_t)^2 = dt$, we have

$$(dS_t)^2 = \sigma^2 S_t^2 dt.$$

Substitution gives

$$d\log S_t = \left(\mu - \frac{1}{2}\sigma^2\right) dt + \sigma dW_t.$$

This explains the $-\frac{1}{2}\sigma^2$ correction in the simulation formula. It is not an arbitrary adjustment; it comes from stochastic calculus.

4.9 Machine-learning regressors and recursive features

The Ridge, Random Forest and Gradient Boosting models use lagged returns, rolling return moments, lagged price levels and time features. Abstractly, they estimate

$$\hat{r}_{t+1} = f(X_t),$$

where X_t contains only information known by time t. Ridge imposes a linear form with an L_2 penalty:

$$\min_{\beta} \sum_{t} (r_{t+1} - X_t\beta)^2 + \lambda\|\beta\|_2^2.$$

Random Forest averages many decision trees to reduce variance. Gradient Boosting builds trees sequentially, with each new tree trying to correct the residuals of the previous ensemble.

The key modelling discipline is recursion. During the test period, the model cannot use actual future returns as lagged inputs. Instead, after predicting $\hat{r}_{T+1}$, it constructs $\hat{S}_{T+1}$ and then uses this predicted value when building features for $T + 2$. This prevents look-ahead bias.

5 Forecasting results and inference

5.1 Accuracy by RMSE

RMSE is

$$RMSE = \sqrt{\frac{1}{h}\sum_{k=1}^{h}(S_{T+k} - \hat{S}_{T+k})^2}.$$

It penalises large price errors more strongly than MAE.

Table 3: Per-hour RMSE on out-of-sample price levels

Model	CVX	NVDA	PG
Naive	2.097	3.933	2.771
ARIMA	2.097	3.933	2.760
GBM	3.484	2.298	3.891
Drift	3.546	2.334	3.926
Ensemble	3.124	2.641	3.236
RandomForest	6.462	3.092	3.242
Ridge	8.564	5.429	3.218
GradientBoosting	10.510	4.309	3.275

CVX is best forecast by Naive and ARIMA$(0, 1, 0)$, which are effectively the same random-walk forecast. This is not because they are mathematically richer. It is because the test period was nearly flat. Trend-following models extrapolated the in-sample decline and were punished when the trend stopped.

NVDA is best forecast by GBM. This makes mathematical sense because the GBM drift term carries the positive in-sample log-return into the test period. The result is not proof that equity prices are truly GBM. It says that, for this short window, a drifted lognormal path was a useful approximation.

PG is best forecast by ARIMA$(2, 1, 2)$. This is the most interesting forecasting result because ARIMA collapses to random walk for CVX and NVDA but not for PG. The lower-volatility staple contains enough serial structure for a non-trivial autoregressive and moving-average specification to help.

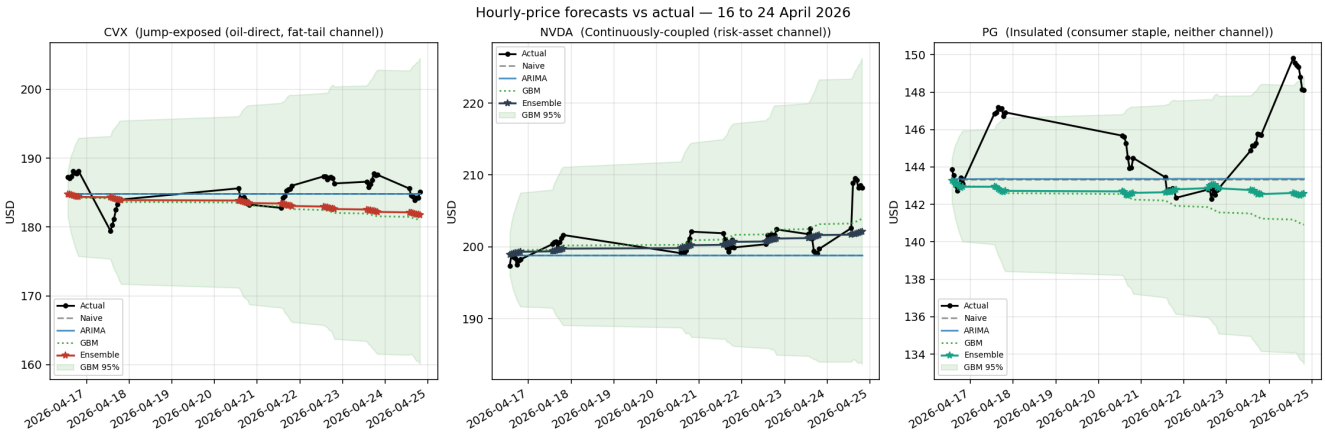


Figure 3: Out-of-sample forecast comparison across CVX, NVDA and PG.

5.2 How to read RMSE, MAE, MAPE and directional accuracy together

RMSE is sensitive to large errors because it squares them. MAE is more linear:

$$MAE = \frac{1}{h}\sum_{k=1}^{h} |S_{T+k} - \hat{S}_{T+k}|.$$

MAPE expresses the error as a percentage of the actual price:

$$MAPE = \frac{100}{h}\sum_{k=1}^{h} \left| \frac{S_{T+k} - \hat{S}_{T+k}}{S_{T+k}} \right|.$$

Directional accuracy ignores magnitude and only checks the sign. Therefore, a model can win RMSE and lose directional accuracy. CVX shows exactly this: the flat random-walk forecast stays close to the level, but it predicts no hour-to-hour movement.

A good viva explanation is: RMSE answers "how far was the price path?" Directional accuracy answers "did the model usually guess the next direction?" Period-mean error answers "was the forecast biased upward or downward?" These are different questions.

5.3 Directional accuracy and period mean

Directional accuracy measures whether the predicted sign of the hourly price change matches the realised sign:

$$DA = \frac{1}{h-1}\sum_{k=2}^{h} \mathbf{1}\left[\operatorname{sgn}(S_{T+k} - S_{T+k-1}) = \operatorname{sgn}(\hat{S}_{T+k} - \hat{S}_{T+k-1})\right].$$

Table 4: Directional accuracy

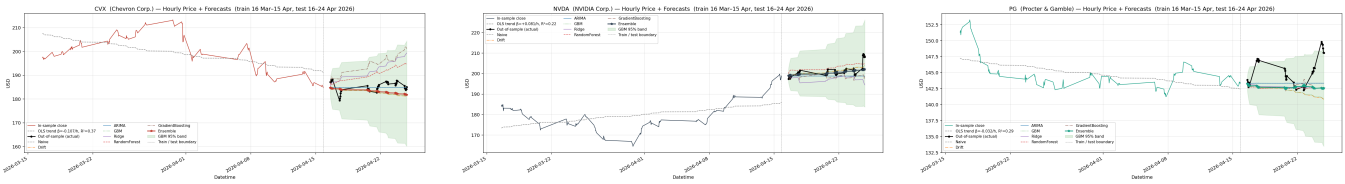
Model	CVX	NVDA	PG
Naive	0.00	0.00	0.00
ARIMA	0.00	0.00	0.54
GBM	0.42	0.56	0.54
Drift	0.42	0.56	0.54
Ensemble	0.42	0.56	0.52
RandomForest	0.35	0.54	0.56
Ridge	0.58	0.50	0.50
GradientBoosting	0.56	0.65	0.56

A flat forecast can win RMSE but lose direction completely. This is why CVX's Naive/ARIMA forecasts have zero directional accuracy even though they have the best RMSE. Ridge and Gradient Boosting do better at signs for CVX, but they are less useful for price-level accuracy.

Table 5: Predicted period-mean price versus realised mean price

Series/model	CVX	NVDA	PG
Actual	185.31	201.33	144.99
Naive	184.83	198.87	143.35
ARIMA	184.83	198.87	143.36
GBM	182.82	201.36	142.07
Drift	182.78	201.29	142.04
Ensemble	183.23	200.50	142.76
RandomForest	190.82	203.12	142.68
Ridge	192.66	197.48	142.74
GradientBoosting	194.78	198.49	142.84

The period-mean table helps separate level bias from path error. NVDA's GBM period mean is almost exactly equal to the realised mean. CVX's ML models overshoot upward, while drift and GBM undershoot because they carry the training decline too far. PG is consistently underpredicted, which means most models captured the level but missed the upward move in the test period.



(a) CVX **(b)** NVDA **(c)** PG

Figure 4: Individual forecast panels with training path, test path, model forecasts and GBM uncertainty band.

6 War-exposure model

6.1 Why one correlation is not enough

A single Pearson correlation at one frequency asks a narrow question: did two return series move together in the same bars? That is not the same as asking whether a company is exposed to war risk. Exposure can arrive continuously, overnight or as a jump.

Pearson correlation is

$$\rho_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y},$$

and OLS beta in the regression $r_t^{stock} = \alpha + \beta r_t^{proxy} + \varepsilon_t$ is

$$\beta = \frac{\mathrm{Cov}(r^{stock}, r^{proxy})}{\mathrm{Var}(r^{proxy})}.$$

Correlation measures standardised co-movement. Beta measures economic sensitivity per 1% proxy move. Both matter.

6.2 Covariance, beta and p-values

The covariance between stock return X and proxy return Y is

$$\mathrm{Cov}(X,Y) = E[(X - E[X])(Y - E[Y])].$$

If this is positive, the variables tend to move in the same direction. If it is negative, they tend to move in opposite directions. Pearson correlation normalises covariance by both standard deviations, so it lies between -1 and 1.

OLS beta is different. In the regression

$$r_t^{stock} = \alpha + \beta r_t^{proxy} + \varepsilon_t,$$

β measures the expected stock return response to a one-unit proxy return. For example, a beta of 0.347 in the overnight Brent-CVX regression means a 1% overnight Brent move is associated with roughly a 0.347% CVX open-gap move, under the fitted linear approximation.

The p-value tests whether the observed relationship could plausibly arise if the true relationship were zero. For Pearson correlation, the test statistic is

$$t = \rho\sqrt{\frac{n-2}{1-\rho^2}},$$

with $n-2$ degrees of freedom under the null. This is why sample size matters: a large correlation with only a few observations can still be uncertain, while a smaller correlation with many observations can be statistically meaningful.

6.3 Lead-lag logic

The lead-lag check shifts Brent returns relative to stock returns:

$$\rho(k) = \mathrm{Corr}(r_t^{stock}, r_{t-k}^{Brent}).$$

If $k > 0$, Brent is allowed to lead the stock. If $k < 0$, the stock leads Brent. A strong positive value at $k = 1$ would suggest that Brent movement one hour earlier is associated with stock movement now. In this report, the most important relationship is not a small intraday lag. It is the overnight channel, where the market-close mismatch is economically meaningful.

6.4 Channel A: continuous intraday co-movement

Table 6: Hourly intraday Pearson correlation with oil proxies, with p-values in parentheses

Proxy	CVX	NVDA	PG
BRENT	0.086 (0.291)	-0.199 (0.013)	-0.031 (0.706)
WTI	0.156 (0.054)	-0.199 (0.014)	-0.002 (0.979)

NVDA is the clearest Channel A name. Its hourly correlations with Brent and WTI are about -0.20 and statistically significant. This negative sign matches the risk-asset story: oil up can mean inflation pressure, uncertainty and risk-off pricing, which can hurt high-beta growth stocks.

CVX has positive oil betas, but weaker intraday correlations. This does not disprove oil exposure. It suggests that the exposure is not evenly distributed across every market-hour bar.

6.5 Channel B: overnight oil-direct transmission

For each trading day, the stock open-gap return is compared with the oil return accumulated while the stock market was closed. This isolates a specific mechanism:

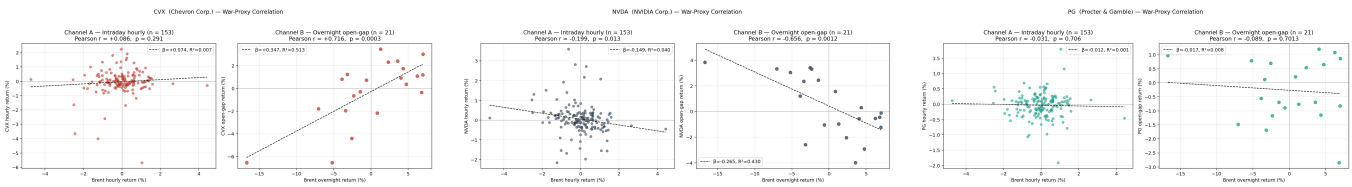
$$r_d^{open} = \frac{S_d^{first\,bar}}{S_{d-1}^{prev\,close}} - 1.$$

Table 7: Overnight/open-gap Pearson correlation with oil proxies, with p-values in parentheses

Proxy	CVX	NVDA	PG
BRENT	0.716 (0.0003)	-0.656 (0.0012)	-0.089 (0.7013)
WTI	0.696 (0.0005)	-0.655 (0.0013)	-0.131 (0.5713)

This is the strongest CVX result. Brent overnight return has a $+0.716$ correlation with CVX open-gap return and explains roughly 51% of its variance. In economic terms, CVX absorbs overnight oil news at the next open. That is exactly the channel an oil major should display.

NVDA also has a large overnight correlation, but it is negative. This confirms that NVDA is exposed through risk sentiment, not oil revenue. PG remains close to zero, which strengthens the control-group interpretation.



(a) CVX **(b)** NVDA **(c)** PG

Figure 5: Intraday and overnight Brent scatter plots. The CVX overnight panel is the cleanest oil-direct relationship.

6.6 Why the three-channel design is mathematically safer

A single statistic can be fragile. The three-channel design reduces that fragility by asking three different mathematical questions:

1. Channel A asks whether stock and proxy returns co-move at the same hourly frequency.
2. Channel B asks whether oil information accumulated outside equity-market hours appears in the next stock open.
3. Channel C asks whether labelled geopolitical event days produce abnormal jumps or asymmetric responses.

This decomposition matters because exposure can be real even when one channel is weak. CVX has modest intraday Pearson correlation, but a very strong overnight Brent relationship. Therefore, a report that used only Channel A would understate CVX's oil exposure.

6.7 Channel C: jump and event exposure

The event study uses five escalation or de-escalation dates. The point-biserial correlation is

$$r_{pb} = \frac{\bar{r}_{esc} - \bar{r}_{de}}{s_r}\sqrt{\frac{n_1 n_0}{n^2}}.$$

It compares the average return on escalation events with the average return on de-escalation events.

Table 8: Event-study and jump metrics

Ticker	Esc. mean (%)	De-esc. mean (%)	r_{pb}	Max $\lvert r\rvert$ (%)	Ex. kurt.
CVX	0.30	-0.27	0.27	1.83	14.31
NVDA	-0.46	2.95	-0.69	5.65	5.03
PG	-1.28	-0.61	-0.30	3.16	5.42

CVX has positive escalation sensitivity and the highest excess kurtosis. NVDA has the strongest point-biserial magnitude and the largest event-day absolute move, driven by its peace-plan rally. PG again has no clean oil-war channel.

The important inference is that jump risk and continuous co-movement are not the same thing. CVX's distribution is full of jump risk even when its intraday Pearson correlation looks modest.

7 Composite exposure score

7.1 Why z-scores are used before averaging

The composite score combines quantities that do not naturally share units: correlations, betas, R^2, kurtosis and event-return magnitudes. Directly averaging them would be mathematically invalid because a beta and a kurtosis value are measured on different scales. Z-scoring solves this by converting every input into a relative cross-sectional position:

$$Z_{i,j} = \frac{x_{i,j} - \bar{x}_j}{s_j}.$$

A positive z-score means the stock is above the three-stock average for that metric. A negative z-score means it is below average. After this transformation, each metric is unit-free and can be averaged within a channel.

The trade-off is that the score is relative, not absolute. NVDA's first rank means it has the strongest combined score among CVX, NVDA and PG. It does not mean NVDA is the most oil-direct stock in the economic world. The channel-level interpretation is therefore more important than the final rank alone. To compare the three channels fairly, each input is converted to a cross-sectional z-score. Within each channel, the z-scores are averaged. The overall score is the average across channels:

$$Z_{i,j} = \frac{x_{i,j} - \bar{x}_j}{s_j}, \qquad Score_i = \frac{1}{3}\left(\bar{Z}_{i,A} + \bar{Z}_{i,B} + \bar{Z}_{i,C}\right).$$

Table 9: Composite war-exposure score by channel

Ticker	A: intraday	B: overnight	C: jump	Composite	Rank
CVX	0.205	0.706	-0.123	0.263	2
NVDA	0.843	0.435	0.541	0.606	1
PG	-1.048	-1.141	-0.419	-0.869	3

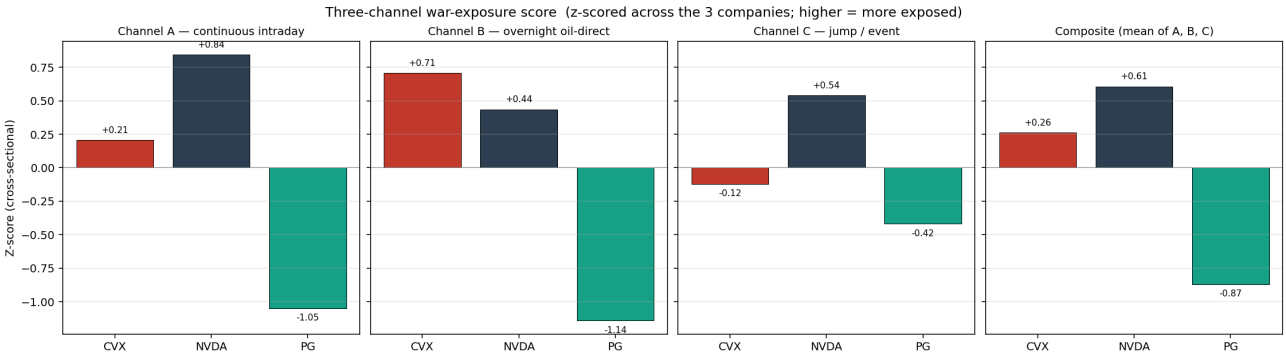


Figure 6: Three-channel composite exposure score.

The composite ranks NVDA first, CVX second and PG third. This may look counterintuitive at first because CVX is the oil company. The resolution is that the composite is broad: it rewards continuous market-hour co-movement and event-day magnitude, where NVDA is very strong. The cleanest oil-direct measure, however, is Channel B, and CVX wins that decisively.

Final exposure verdict. CVX is not less oil-exposed than NVDA. It is more oil-direct. NVDA is more continuously sensitive to geopolitical risk sentiment. PG is the insulated benchmark. The difference is not the amount of risk alone, but the channel through which risk enters the price.

8 Limitations and improvements

8.1 Short sample length

The sample has only 153 hourly returns and 21 daily overnight observations. That is enough to demonstrate mechanisms, but not enough to claim stable long-run parameters. A longer rolling window would narrow confidence intervals and reduce event-selection sensitivity.

8.2 Constant volatility in GBM

GBM assumes constant volatility. The excess kurtosis values show that this is too simple, especially for CVX. A better model would use GARCH or a jump-diffusion process:

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t + (J-1)dN_t,$$

where N_t is a Poisson jump process and J is the random jump multiplier. This would separate normal diffusion risk from discrete geopolitical shock risk.

8.3 Exogenous information

The current forecasting models are mostly univariate. A stronger design would use ARIMAX or VAR models with Brent, WTI, sector ETFs, interest rates and volatility indices. That should help most for CVX, because the strongest evidence says CVX's price absorbs oil information overnight.

8.4 Inference robustness

Newey-West standard errors would help because high-frequency returns can have heteroskedasticity and weak autocorrelation. Bootstrap confidence intervals would also be useful for the composite score because it is built from a small cross-section and a short sample.

9 Conclusion

This report shows why finance modelling needs both mathematics and market structure. A single daily return, a single Pearson correlation or a single forecasting metric can all point in the wrong direction.

CVX looks like a jump-sensitive oil major. Its exposure is concentrated in the overnight open gap and in the fat left tail. NVDA looks like a high-beta risk asset. Its exposure is continuous and sentiment-driven. PG looks like the defensive control, with low volatility and weak proxy relationships.

The forecasting results tell the same story from another angle. Random walk models win when the test path is

flat. GBM wins when drift persists. ARIMA wins when low-volatility serial structure is present. The best model is therefore not universal. It depends on the stock's statistical texture and the regime being forecast.

A Files used in this report

File pattern	Purpose
`task2_pipeline.py`	Single analysis pipeline used as the source of truth.
`CVX.csv`, `NVDA.csv`, `PG.csv`	Cached hourly OHLC data for the three companies.
`BZ_F.csv`, `CL_F.csv`, `LMT.csv`	Hourly proxy data for Brent, WTI and Lockheed Martin.
`{TICKER}_forecast_metrics.csv`	RMSE, MAE, MAPE, directional accuracy and period-mean forecast metrics.
`{TICKER}_war_correlation.csv`	Hourly and daily Pearson, Spearman and beta results against proxies.
`{TICKER}_overnight_correlation.csv`	Open-gap correlations with overnight oil returns.
`{TICKER}_event_study.csv`	Event-day returns for escalation and de-escalation dates.
`comparison_*.csv`	Cross-company summary, forecast and war-exposure tables.
`*.png`	Distribution, forecast, scatter and comparison figures included in the report.

B Reproducibility note

The report is generated from the supplied CSV and PNG outputs. To reproduce the analysis tables and figures from raw cached data, run:

```
python task2_pipeline.py
```